

RUT Tier 4 Sprint A+B+C Replication Packet

Background, Growth, and Hardened Pilot E_G Tests

Prepared for Mark Van Dyk / Craft Conscious LLC - Packet v1 - 2026-05-13

Executive Summary

This packet collects the outputs from RUT Sprint A, Sprint B, Sprint C, and Sprint C-H. The strongest current result is a hardened pilot E_G / lensing result: fixed neutral-lensing RUT beats LCDM across the full diagonal E_G compilation, one-per-family subset, family-weighted subset, and leave-one-point-out jackknife checks. This is not a confirmed detection; it remains a pilot result pending covariance-clean replication.

Conservative status: Tier 4 pipeline with weak but hardened Tier 4-style evidence emerging in the E_G / lensing sector.

Key Model Form

$$B(a) = 1/2 [1 + \tanh((a - a_t) / \Delta a)]$$

$$\mu(a) = 1 + \mu_0 B(a)$$

$$\Sigma(a) = 1 + \lambda_\Sigma \mu_0 B(a)$$

The best fixed E_G-lane model is neutral lensing: $\lambda_\Sigma = 0$, so $\Sigma(a) = 1$.

Key Results

Stage	Check	Headline result	Read
Sprint A	Background real data	Delta BIC about +3.7	LCDM preferred
Sprint B	Joint background + growth	Delta BIC about +3.63; μ_0 about 0.173	Mild growth hint, not BIC-supported
Sprint C	Pilot multi-point E_G	RUT neutral Delta BIC about -2.108	First weak RUT advantage
Sprint C-H	One-per-family E_G	RUT neutral Delta BIC about -1.926	Survives family reduction
Sprint C-H	Family-weighted E_G	RUT neutral Delta BIC about -1.963	Survives weighting
Sprint C-H	Leave-one-point-out	Win fraction 1.0; worst Delta BIC about -1.266	No single point carries result

Caveats

This is not a confirmed detection. The E_G compilation is diagonal-error only, some measurements may overlap in survey data or methodology, covariance matrices were not fully incorporated, and the result should be tested on a publication-grade covariance-clean E_G / lensing subset.

Replication

Run: `python scripts/verify_results.py`

The script verifies the headline Sprint C-H claims from the stored CSV outputs and prints PASS if the packet reproduces the key claims.

Recommended Upload Locations

GitHub: [results/sprint_c_hardened_EG_replication_packet/](#)

OSF: RUT Results / Sprint C-H - E_G Independence Hardening

Suggested commit message: Add Sprint C-H hardened E_G replication packet